AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_C5FE13BA1C
Taxa	Stegosaurus
Clade	Ornithischian
PBDB Records	86
Mass (regression)	3576.6 kg (Campione & Evans 2012)
Mass (volumetric)	3441.3 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	3.622 m³
Validity Score	1.0000 / 1.0000
TCI	0.6610 (partial)
Axiom I Entropy	0.9753
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.5108
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.259, 1.143, 0.0]
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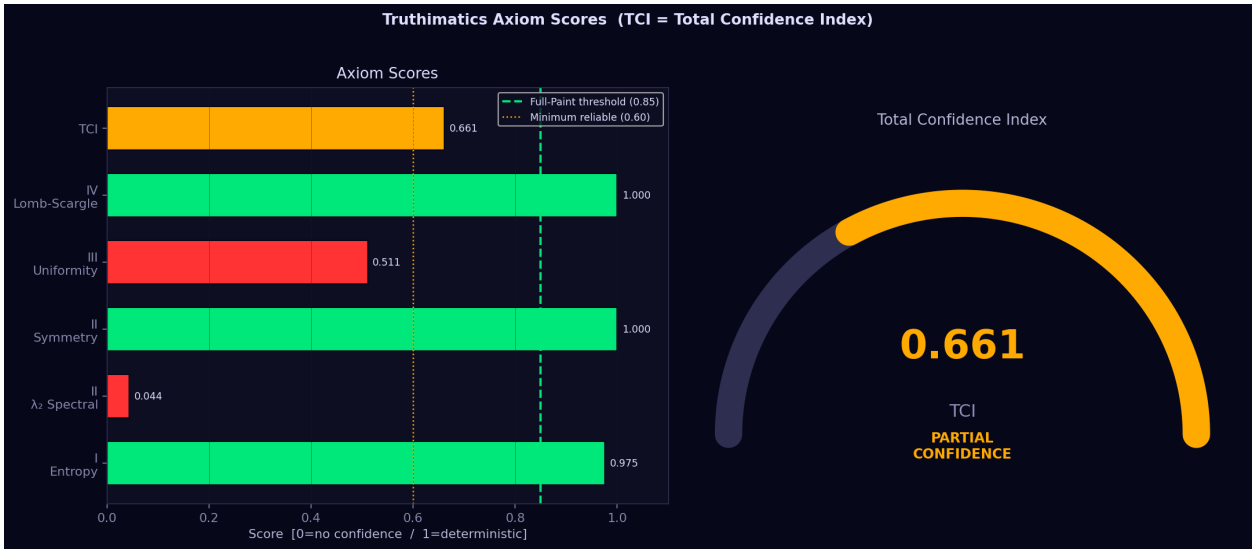
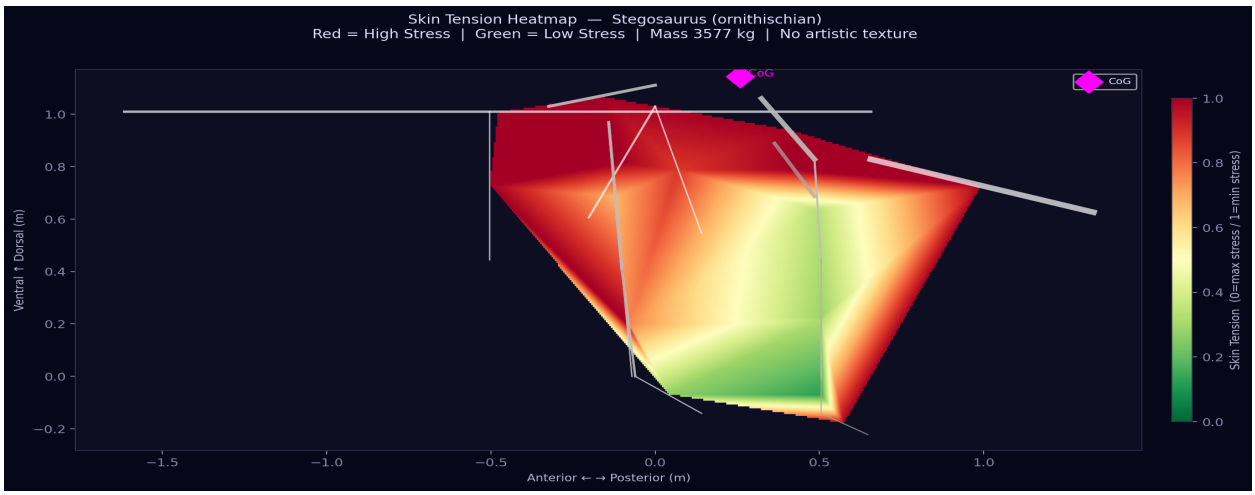
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



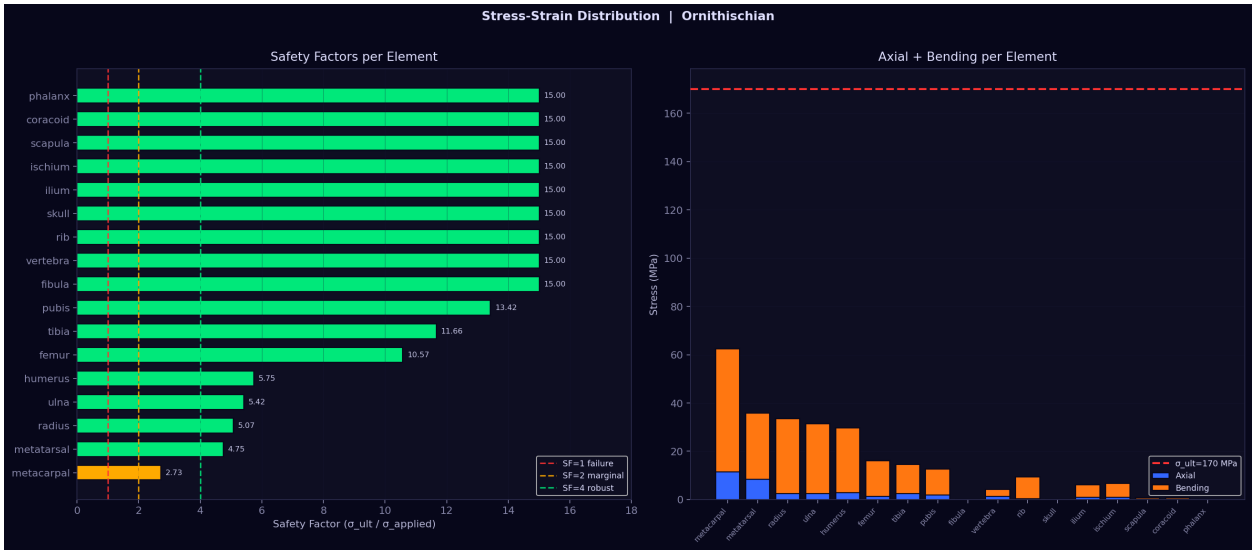
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: quadruped (F=Mg/4). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

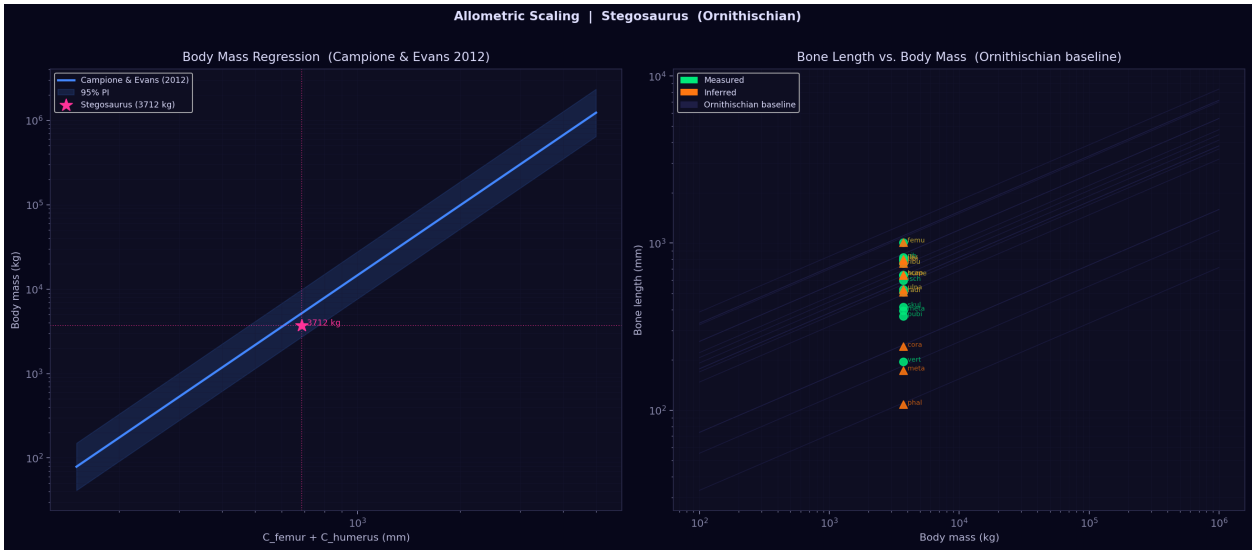


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.255	0.465	0.719	15.000	OK
femur	1.387	14.692	16.079	10.573	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.806	26.782	29.588	5.746	OK
ilium	0.946	5.243	6.189	15.000	OK
ischium	0.863	5.744	6.608	15.000	OK
metacarpal	11.485	50.863	62.348	2.727	WARNING
metatarsal	8.405	27.424	35.829	4.745	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.826	10.844	12.671	13.417	OK
radius	2.537	31.007	33.544	5.068	OK
rib	0.262	9.118	9.380	15.000	OK
scapula	0.144	0.531	0.675	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	2.366	12.213	14.579	11.660	OK
ulna	2.365	29.003	31.368	5.420	OK
vertebra	1.387	2.837	4.224	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.19 m	Femur + tibia + metatarsal
Estimated Speed	13.8 km/h (3.84 m/s)	Gait: Walk | Fr=0.686
Stride Length	4.487 m	Alexander (1976) / Thulborn (1990)
Bite Force	2178.5 N (222.1 kgf)	Ornithischian regression (Norman & Weishampel 1991)
Skull Length	415.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3981	Circumference / length (dimensionless)
Limb Strength Index	4.3e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	9.09 m	Allometric from femur length
Est. Neck Length	1.0 m	Allometric from femur length
Est. Tail Length	2.54 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	240.3	176.7	3006.30	3752.8	Constraint	0.40	15.00
femur	1010.0	128.0	6628.29	8274.1	Measured	0.88	10.57
femur_right	1010.0	128.0	6628.29	8274.1	Symmetry	0.84	10.57
fibula	755.0	40.0	341.55	358.2	Measured	0.88	15.00
fibula_right	755.0	40.0	341.55	358.2	Symmetry	0.84	15.00
humerus	640.0	90.0	2076.47	2592.1	Measured	0.88	5.75
humerus_right	640.0	90.0	2076.47	2592.1	Symmetry	0.84	5.75
ilium	800.0	155.0	7698.63	9610.2	Measured	0.88	15.00
ilium_right	800.0	155.0	7698.63	9610.2	Symmetry	0.84	15.00
ischium	595.0	96.0	2196.44	2741.8	Measured	0.88	15.00
metacarpal	172.8	43.1	47.95	39.4	Constraint	0.65	2.73
metatarsal	395.0	52.0	427.82	534.1	Measured	0.88	4.75
phalanx	108.1	35.3	16.30	12.6	Constraint	0.40	15.00
pubis	365.0	66.0	636.86	795.0	Measured	0.88	13.42
radius	510.0	56.0	640.63	799.7	Measured	0.88	5.07
radius_right	510.0	56.0	640.63	799.7	Symmetry	0.84	5.07
rib	820.0	38.0	474.29	592.1	Measured	0.88	15.00
scapula	645.0	235.0	14267.75	17810.4	Measured	0.88	15.00
scapula_right	645.0	235.0	14267.75	17810.4	Symmetry	0.84	15.00
skull	415.0	308.0	15769.20	19684.7	Measured	0.88	15.00
tibia	785.0	98.0	3019.83	3769.6	Measured	0.88	11.66
tibia_right	785.0	98.0	3019.83	3769.6	Symmetry	0.84	11.66
ulna	530.0	58.0	714.15	891.5	Measured	0.88	5.42
ulna_right	530.0	58.0	714.15	891.5	Symmetry	0.84	5.42
vertebra	195.0	128.0	1279.72	1597.5	Measured	0.88	15.00